HPC Cluster 101

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Cincinnati Children's
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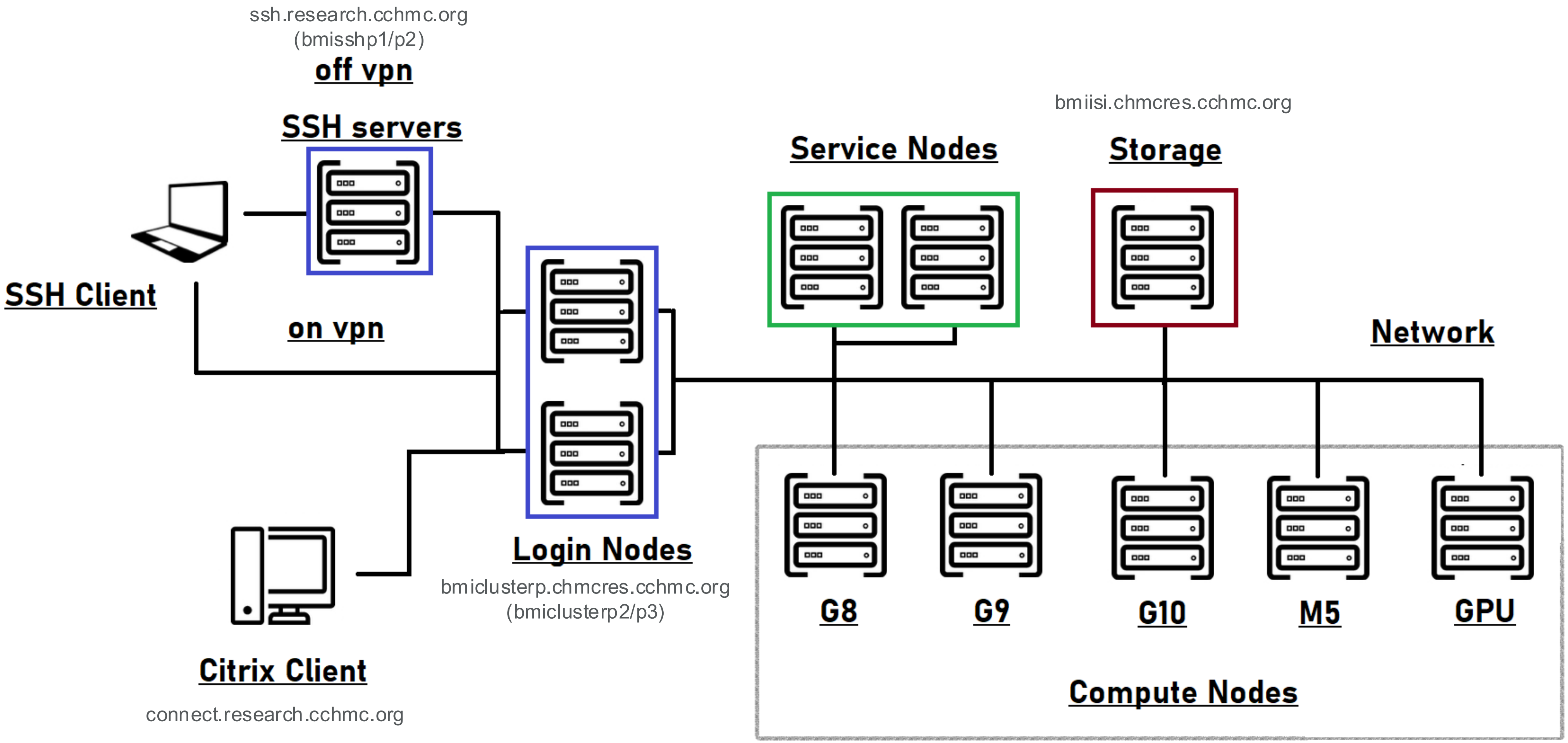
Overview

- HPC cluster at a glance
- How to access
- Logging in
- Storage & Data Transfer
- Software Modules
- Running & troubleshooting jobs
- Cluster policies
- New features
- Demo!

HPC Topology



Cluster specifications

- 2 login nodes
- 2 service nodes
- 83 compute nodes (numbers are approx)
  - 26 HP 460g8 with 16 cores and 256GB RAM
  - 23 HP 460g9 with 16 cores and 256GB RAM
  - 5 HP 460g10 with 28 cores and 256GB RAM
  - 10 Cisco M5 with 24 cores and 256GB RAM
  - 4 Cisco M5 with 56 cores and 768GB RAM
  - 8 Cisco M6 with 32 cores and 512GB RAM (new)
  - 5 Dell R740 with 2 x Nvidia V100 GPUs (16 cores, 384GB RAM)
  - 2 Dell XE8545 with 4 x Nvidia A100 GPUs (128 cores, 512GB RAM)
- 10GbE interconnect
- Dell/EMC Isilon storage for home, scratch, data volumes

FAQ #1: How to get access

- E-mail help-cluster@bmi.cchmc.org
- Accounts can typically be created within a day
- You'll get an email from us with instructions and a link to our website which has more info: https://hpc.research.cchmc.org

Logging in

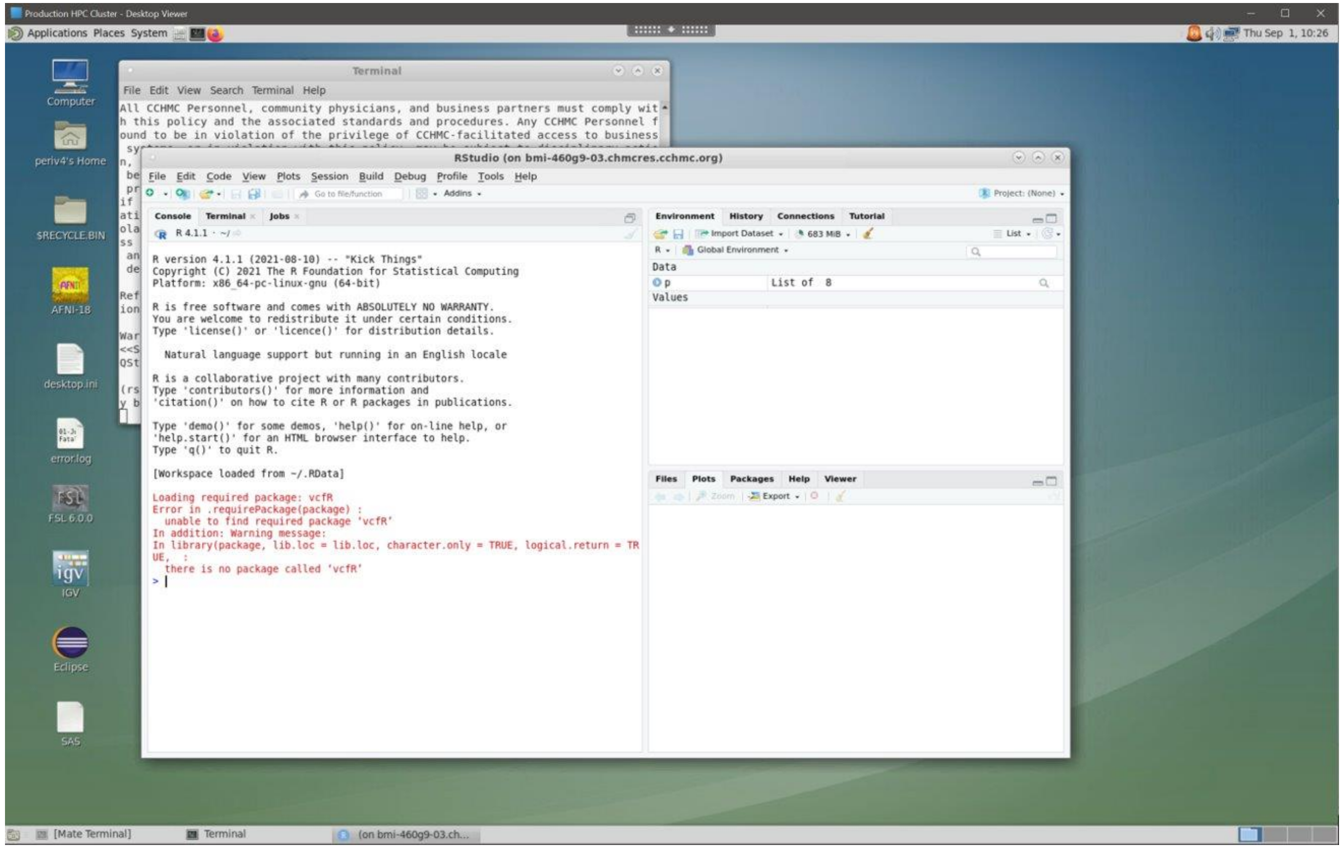
- Two ways: Citrix (graphical) and SSH (command line)
- Login nodes – bmiclusterp2/p3
  - These are NOT for running jobs!
  - Any process consuming excessive resources will be killed
  - Light usage is ok: editing code, file management, etc…
  - Gray areas: tar, gzip, find, grep (depends on the size)
  - tmux and screen are your friends
  - *FAQ #2*: Why was my job killed? Ans: did you run it on the login node by mistake?

Logging in - SSH

- Use your favorite ssh client
  - Windows: PuTTY, VSCode, Powershell?
  - Mac and Linux: terminal
- `> ssh <username>@bmiclusterp.chmcres.cchmc.org`
  - You will land on bmiclusterp2
- Off site? Use VPN or ssh gateways (ssh.research.cchmc.org)
- Advanced: If you have an X server, you can turn on X forwarding with -X or -Y options

Logging in - Citrix

- Point your browser to connect.research.cchmc.org and log in with CCHMC credentials
- Instructions are on our website
- Provides full desktop to use graphical applications (e.g. Rstudio, Matlab)
  - Or just a more familiar interface if you are not comfortable with CLI
- You will land on either bmiclusterp2 or p3



Storage

- Every user gets (free):
  - 100GB home directory
  - Up to 5TB scratch directory (more on this later)
- Additional storage for a fee (request via the “store”):
  - Additional data volumes, typically created for individual labs or groups; mounted under /data on the cluster
  - ritstore.research.cchmc.org
- RDS6 shares can also be accessed on the cluster
  - `rds connect <share name>`
  - `rds disconnect <share name>`
  - `rds credentials` #Creates an encrypted file for passwordless use
- Some reference databases & genomes can be found under /database on the cluster (module specific)

Data Transfer

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- Primary method: mount your home or scratch on your laptop/desktop
  - Almost all Isilon (bmiisi) shares are available over both NFS and SMB
  - Windows: \\bmiisi.chmcres.cchmc.org\home\<username>
  - Mac: smb://bmiisi.chmcres.cchmc.org/home/<username>
  - Linux: `mount.cifs`
- Other methods
  - File transfer tools: scp, rsync, robocopy, WinSCP
  - Connect via bmiclusterp (onsite/VPN) or ssh.research.cchmc.org (offsite)
  - Data volumes are also mounted on ssh.research.cchmc.org, making it a good data transfer node
  - Sometimes data transfers can utilize excessive CPU; may need to launch an interactive job to use a compute node

Outbound access

- Downloading packages or data sets from cluster is possible via proxy
- Proxy access is NOT enabled by default – must request access via help-cluster ticket
  - *FAQ #3*: “I’m getting 407 error when running my script / downloading a package”
- Run “proxy_on” at command line to set environment variables (e.g. http_proxy, https_proxy)
  - It will prompt for username and password
  - If your password has certain special characters it may not work – 2 options:
    - Change your password
    - Use URL encoding (e.g. %## instead of x)
- We do maintain a white list of sites which are accessible without needing proxy! (Cisco Umbrella – DNS)
  - `iswhitelisted <URL>` - verify if a site is in the whitelist

Software modules

- *FAQ #4*: "Is package xyz available on the cluster?"
  - `module avail <pkg name>`
- Using installed packages
  - `module load <pkg name>/<version>`
  - Modifies environment variables (e.g. $PATH) to use correct binaries and libraries
  - `module list` # list all loaded packages in env
  - `module unload <pkg name>` # unload package from env
  - `module purge` # unload all packages from env
- Package not listed?
  - We can install it for you! Request via email to help-cluster
  - Sometimes you can install it yourself in your home dir
    - However, keep in mind your home dir only has 100GB

Conda, Pip, and R

- Some software has a preferred installation method
- Conda
  - `module load anaconda3`
  - `conda list`
  - `source activate <env name>`
  - DO NOT USE conda init
    - Edits .bashrc which can cause issues with Citrix and other packages
- Pip
  - each version of python has its own set of packages
  - Can be installed centrally by admin
- R libraries
  - each version of R has its own set of packages
  - you can install your own using home dir
- We maintain some (or global) environments centrally, but you can also build your own in your home dir

Submitting Jobs

- IBM LSF Scheduler
  - Ensures fair sharing of resources among all users
  - `bsub` #job submission command
    - -M <num> # memory
    - -n <num> # cores
    - -W <time> # wall time
    - -Is # interactive
    - -q <queue name>
  - batch example
    - `bsub < job.txt`
  - interactive example
    - `bsub -W 2:00 -M 32000 -n 2 -Is /bin/bash`
- Queues – groups of nodes with a set of policies, users, priority
  - ”normal” : this is default
  - project specific
  - GPU and Docker queues

job.txt

```
#BSUB -W 2:00
#BSUB -n 2
#BSUB -M 32000
#BSUB -e <some directory>/%J.err
#BSUB -o <some directory>/%J.out

module load R
cd ~
# execute program
R CMD BATCH runme.R
```

Specialty jobs

- GPU
  - Queues: gpu-v100, gpu-a100
  - `bsub -q gpu-v100 -gpu "num=1" ...`
  - CUDA modules available
- Docker
  - `bsub -q docker ...`
  - Use scratch space and not /data shares
  - Add sticky bit to any output directory
    - `chmod +t /scratch/<dest_dir>`

Troubleshooting jobs – bhist/bjobs

bjobs output

```
JOBID   USER    STAT  QUEUE      FROM_HOST   EXEC_HOST   JOB_NAME   SUBMIT_TIME
93457   hee9ke  RUN   gpu-v100   bmi-200m6-0 2*bmi-r740- *_stain.py May 31 16:59
94569   raw6jg  RUN   gpu-v100   bmiclusterp 4*bmi-r740- $SAMPLE    Jun  1 10:43
95415   srv-sab RUN   web2       bmiwebe3    bmi-460g8-4 SABLE      Jun  1 12:58
81996   iwap3b  USUSP normal     bmiclusterp 2*bmi-460g9 *multiqc . May 23 16:20
79050   wan5fg  PSUSP gpu-v100   bmi-460g9-2             1037_2000  May 22 16:22
94593   lewt2p  PEND  normal     bmi-200m5-0             *fReli_(5) Jun  1 10:51
94574   lewt2p  PEND  normal     bmi-200m5-0             *Reli_(20) Jun  1 10:50
94576   lewt2p  PEND  normal     bmi-200m5-0             *fReli_(3) Jun  1 10:50
```

Normal job states

Most jobs enter only three states:

Job state	Description
PEND	Waiting in a queue for scheduling and dispatch
RUN	Dispatched to a host and running
DONE	Finished normally with a zero exit value

Suspended job states

If a job is suspended, it has three states:

Job state	Description
PSUSP	Suspended by its owner or the LSF administrator while in PEND state
USUSP	Suspended by its owner or the LSF administrator after being dispatched
SSUSP	Suspended by the LSF system after being dispatched

Exceeding CPU limit

```
Every 2.0s: bjobs -l 72974


Job <72974>, User <colu3p>, Project <default>, Status <EXIT>, Queue <normal>, I
                     nteractive pseudo-terminal shell mode, Job Priority <50>,
                     Command <bash>, Esub <set-defaults dynamic-reject>
Thu May 18 15:56:37: Submitted from host <bmiclusterp2>, CWD <$HOME/trash>, Req
                     uested Resources < rusage[mem=512] order[cpuf:-mem]>, Spec
                     ified Hosts <bmi-200m5-10>;
Thu May 18 15:56:38: Started on <bmi-200m5-10>, Execution Home </users/colu3p>,
                      Execution CWD </users/colu3p/trash>;
Thu May 18 16:10:56: Exited by LSF signal TERM_CPULIMIT. The CPU time used is 2
                     165.9 seconds.
Thu May 18 16:10:56: Completed <exit>; TERM_CPULIMIT: job killed after reaching
                      LSF CPU usage limit.

 CPULIMIT
 36.0 min of bmi-200m5-10
 RUNLIMIT
 60.0 min

 MEMLIMIT
    512 M

 MEMORY USAGE:
 MAX MEM: 132 Mbytes;  AVG MEM: 83 Mbytes; MEM Efficiency: 25.78%

 CPU USAGE:
 CPU PEAK: 7.33 ;  CPU Efficiency: 100.00%

 SCHEDULING PARAMETERS:
           r15s   r1m  r15m   ut      pg    io   ls    it    tmp    swp    mem
 loadSched   -     -     -     -       -     -    -     -     -      -      -
 loadStop    -     -     -     -       -     -    -     -     -      -      -

 RESOURCE REQUIREMENT DETAILS:
 Combined: select[type == any] order[cpuf:-mem] rusage[mem=512.00]
 Effective: select[type == any] order[cpuf:-mem] rusage[mem=512.00]
```

Exceeding Memory limit

```
[colu3p@bmiclusterp2 trash]$ bhist -l 72941


Job <72941>, User <colu3p>, Project <default>, Interactive pseudo-terminal shel
                     l mode, ssh X11 forwarding mode, Command </bin/bash /usr/l
                     ocal/bin/terminal.bat>, Esub <set-defaults dynamic-reject>
Thu May 18 15:27:16: Submitted from host <bmiclusterp2>, to Queue <normal>, CWD
                      <$HOME>, Requested Resources <span[ptile=1] rusage[mem=10
                     24] order[cpuf:-mem]>;
Thu May 18 15:27:17: Dispatched to <bmi-200m5-04>, Effective RES_REQ <select[ty
                     pe == local] order[cpuf:-mem] rusage[mem=1024.00] span[pti
                     le=1] >;
Thu May 18 15:27:18: Starting (Pid 219418);
Thu May 18 15:27:19: Running with execution home </users/colu3p>, Execution CWD
                      </users/colu3p>, Execution Pid <219418>;
Thu May 18 15:44:54: Exited with exit code 143. The CPU time used is 432.7 seco
                     nds;
Thu May 18 15:44:54: Completed <exit>; TERM_MEMLIMIT: job killed after reaching
                      LSF memory usage limit;

 RUNLIMIT
 60.0 min of bmiclusterp2

 MEMLIMIT
      1 G

MEMORY USAGE:
MAX MEM: 1 Gbytes;  AVG MEM: 361 Mbytes; MEM Efficiency: 100.00%

CPU USAGE:
CPU PEAK: 0.68 ;  CPU Efficiency: 68.25%

Summary of time in seconds spent in various states by  Thu May 18 15:44:54
  PEND     PSUSP    RUN      USUSP    SSUSP    UNKWN    TOTAL
  1        0        1057     0        0        0        1058
```

Job Suspended

```
Thu Jun  1 12:49:14: Resource usage collected.
                     The CPU time used is 31774 seconds.
                     MEM: 6.7 Gbytes;  SWAP: 0 Mbytes;  NTHREAD: 18
                     PGID: 6447;  PIDs: 6447 6458 6462 6463 6504 9007 9008
                     9012 9013 9016 9017 9018

SUSPENDING REASONS:
Job was suspended by user;
RUNLIMIT
2400.0 min

MEMLIMIT
  15.6 G

MEMORY USAGE:
MAX MEM: 7.3 Gbytes;  AVG MEM: 6.7 Gbytes; MEM Efficiency: 47.20%

CPU USAGE:
CPU PEAK: 5.84 ;  CPU Efficiency: 100.00%

SCHEDULING PARAMETERS:
          r15s   r1m  r15m   ut      pg    io   ls    it    tmp    swp    mem
loadSched   -     -     -     -       -     -    -     -     -      -      -
loadStop    -     -     -     -       -     -    -     -     -      -      -

RESOURCE REQUIREMENT DETAILS:
Combined: select[type == local] order[cpuf:-mem] rusage[mem=16000.00]
Effective: select[type == local] order[cpuf:-mem] rusage[mem=16000.00]
```

Job troubleshooting

- Look at your job logs for useful info
  - Sometimes this is emailed to you
    - #BSUB options –B and –N

- `bjobs` – Display jobs which are running or in queue
- `bhist` – Display jobs which have completed

  -u <user> Search by userID or "all" for all users
  -q <queue>
  -l Display in a long format
  -m <host> Display jobs dispatched to a specified host
  -n <#> bhist only. Number of log files to search, 0=all
    - bhist searches can take several minutes to run – may need compute node

- `bpeek` – monitor stdin and stderr of a running job
  - -f Follow a running job

- `bhosts` – list compute nodes' status

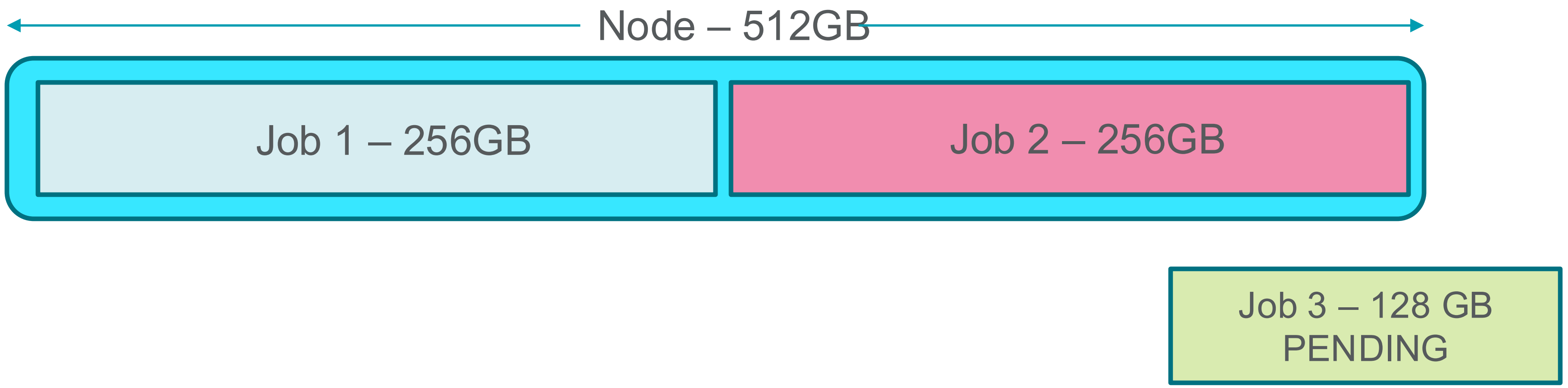
```
[colu3p@bmiclusterp2 ~]$ bjobs
JOBID   USER    STAT  QUEUE      FROM_HOST   EXEC_HOST   JOB_NAME   SUBMIT_TIME
72922   colu3p  RUN   normal     bmiclusters bmi-200m5-0 bash       May 18 15:13
[colu3p@bmiclusterp2 ~]$ bjobs -l

Job <72922>, User <colu3p>, Project <default>, Status <RUN>, Queue <normal>, In
                     teractive pseudo-terminal shell mode, Job Priority <50>, C
                     ommand <bash>, Esub <set-defaults dynamic-reject>
Thu May 18 15:13:36: Submitted from host <bmiclustersvcp1>, CWD <$HOME>, Reques
                     ted Resources < rusage[mem=512] order[cpuf:-mem]>;
Thu May 18 15:13:37: Started on <bmi-200m5-02>, Execution Home </users/colu3p>,
                      Execution CWD </users/colu3p>;
Thu May 18 15:18:12: Resource usage collected.
                     MEM: 4 Mbytes;  SWAP: 0 Mbytes;  NTHREAD: 5
                     PGID: 125403;  PIDs: 125403
                     PGID: 125417;  PIDs: 125417 125419
                     PGID: 125420;  PIDs: 125420

 CPULIMIT
 36.0 min of bmi-200m5-02
 RUNLIMIT
 60.0 min

 MEMLIMIT
    512 M

 MEMORY USAGE:
 MAX MEM: 7 Mbytes;  AVG MEM: 4 Mbytes; MEM Efficiency: 1.37%

 CPU USAGE:
 CPU PEAK: 0.00 ;  CPU Efficiency: 0.00%

 SCHEDULING PARAMETERS:
           r15s   r1m  r15m   ut      pg    io   ls    it    tmp    swp    mem
 loadSched   -     -     -     -       -     -    -     -     -      -      -
 loadStop    -     -     -     -       -     -    -     -     -      -      -

 RESOURCE REQUIREMENT DETAILS:
 Combined: select[type == local] order[cpuf:-mem] rusage[mem=512.00]
 Effective: select[type == local] order[cpuf:-mem] rusage[mem=512.00]
```

Job contention

- We are seeing many jobs which request much more memory than needed
- Can block other jobs from running (example on next slide)
- Please try to request only what you need – let’s be good HPC citizens!

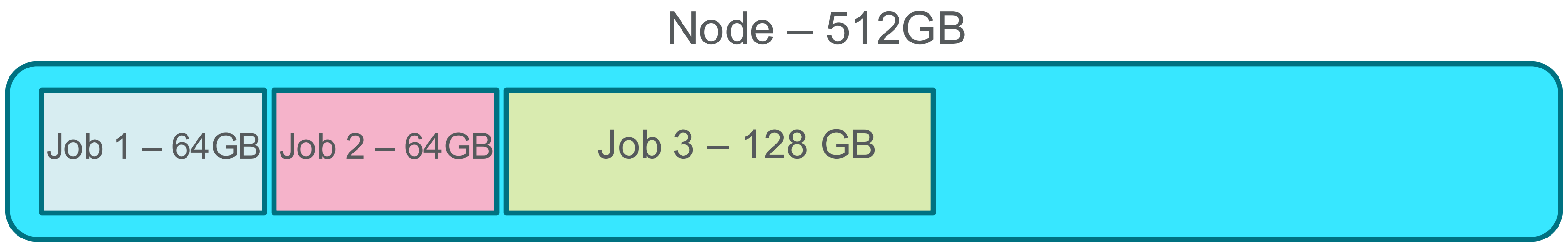
Memory Example 1

- Node has 512GB RAM
- Job 1 is asking for 256GB RAM, is scheduled on node
- Job 2 is asking for 256GB RAM, is scheduled on node
- Both jobs never use more than 32GB RAM, but no other jobs can run because all memory is reserved (LSF must assume your program *could* use more RAM at some point)
- Job 3 is asking for 128GB RAM but must wait until other jobs finish



Memory Example 2

- Node has 512GB RAM
- Job 1 is asking for 64GB RAM, is scheduled on node
- Job 2 is asking for 64GB RAM, is scheduled on node
- Job 3 is asking for 128GB RAM and can be scheduled immediately since there is free RAM



Memory tuning

- How do I know how much memory I need?!
  – Prior experience
  – Size of data set can be a good place to start
    - Reading a 6GB bam file – maybe start with 8GB?
  – There are utilities that can profile your program
  – LSF will tell you! (see previous slides on bjobs and bhist)
- You should overestimate, but don't go crazy!
  – If you program needs 16GB and you request 16GB, there's not much room for error
  – If you need 16GB, requesting 24GB is reasonable
- Max memory limits are ~ 17GB less than system memory
  – `lshosts`

Cluster policies

- Cluster is open to all CCHMC employees
  - Given valid use case; you can't mine bitcoin
  - External users can access as collaborators
- Cluster hours
  - Every user is allocated 10,000 core hours each quarter
    - NOTE: these hours are per **core** and per **job** (ex: bsub –n 4 –W 2:00 … -> will use 8 core hours)
      - *FAQ #5:* "Where did my hours go???"
  - $0.01 per core hour after that, by request
  - *FAQ #6*: "Why is my job stuck in PEND/PSUSP?" (you may not have enough hours to cover the job you are trying to submit)
    - Watch your "reserved" hours
  - Check your balance: `module load gold` then `gbalance -h`
- Job/User limits:
  - 754GB max RAM per job
  - 3.57TB max RAM per user
  - 125 max running jobs/cores per user (can be increased on request)
  - 1250 max pending jobs per user (can be increased on request)

Cluster policies

- Scratch storage - /scratch/<youruserid>
  - Designed to be a workspace for your jobs
  - Ideal for docker jobs
  - Permissions are UNIX based vs DP based for /data shares
  - Baseline limit of 100GB, with a 7 day burst limit of 5TB
  - Check limits with `scratchusage`

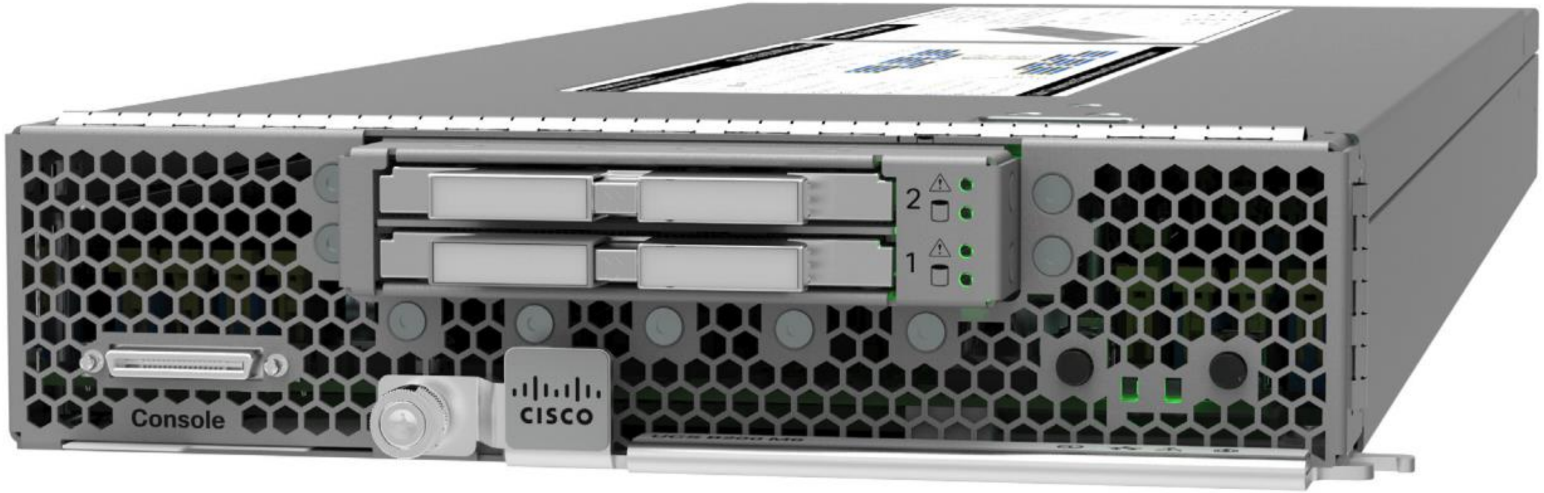
```
$ scratchusage
Scratch Usage: 73GB Hard Limit: 5120GB
100GB Base quota is not currently in violation.
```

Cluster maintenance

- When: every 90 days
- Why: To ensure the latest security and stable releases on the OS are installed
- Impact: Citrix and ssh sessions will be disconnected, including interactive jobs, rstudio and matlab programs
- An email will be sent a week prior, with up to 3 reminders
- It is advisable to plan and save your work during the outage

New HPC nodes

- Intel Xeon Gold 6326 – 2.9 Ghz, 16 cores
- Cisco M6 blade – two CPUs, 512GB RAM



Per user job limits increased

- Default max of 125 cores per user (up from 100)
- Will increase to 150 if monitoring looks ok
- Reminder on how cores are counted:
  - bsub –n 8 #asking for 8 cores
  - If I run 4 of these jobs, I'm using 32 out of 125 max cores

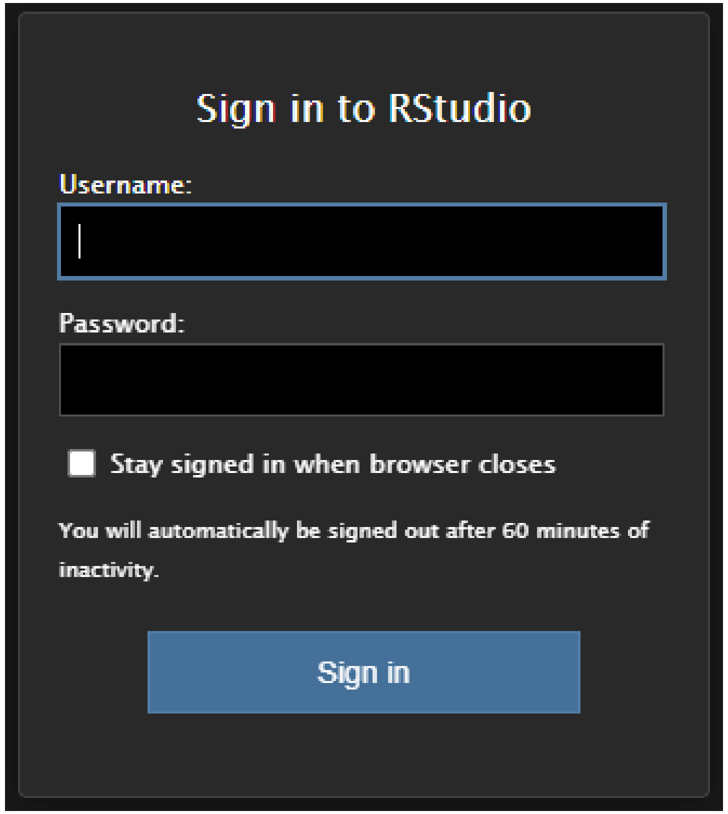
Rstudio in the browser

- Run this command in an interactive session:
  - `/usr/local/bin/rserver.sh`

From your CITRIX session
Open a Firefox/Chromium page on the CITRIX server and point to:
http://bmi-460g8-01.chmcres.cchmc.org:33592

or

From your workstation
ssh -L 8787:bmi-460g8-01.chmcres.cchmc.org:33592 root@bmiclusterp.chmcres.cchmc.org -p 22
Open a Firefox/Chromium page and point to:
http://localhost:8787



Jupyter Notebook

```
$ module load anaconda3
$ source activate myenv
$ jupyter-notebook --ip=0.0.0.0 --port=8888 --no-browser

[I 15:12:12.423 NotebookApp] Loading IPython parallel extension
[I 15:12:13.967 NotebookApp] JupyterLab extension loaded from /usr/local/anaconda3-2020/lib/python3.8/site-packages/jupyterlab
[I 15:12:13.968 NotebookApp] JupyterLab application directory is /usr/local/anaconda3-2020/share/jupyter/lab
[I 15:12:13.972 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 15:12:13.990 NotebookApp]

    To access the notebook, open this file in a browser:
        file:///users/periv4/.local/share/jupyter/runtime/nbserver-24375-open.html
    Or copy and paste one of these URLs:
        http://bmi-460g8-01.chmcres.cchmc.org:8888/?token=7f33467e41fba92d6adde3789b4bdbe7b44f06c22a042b27
```